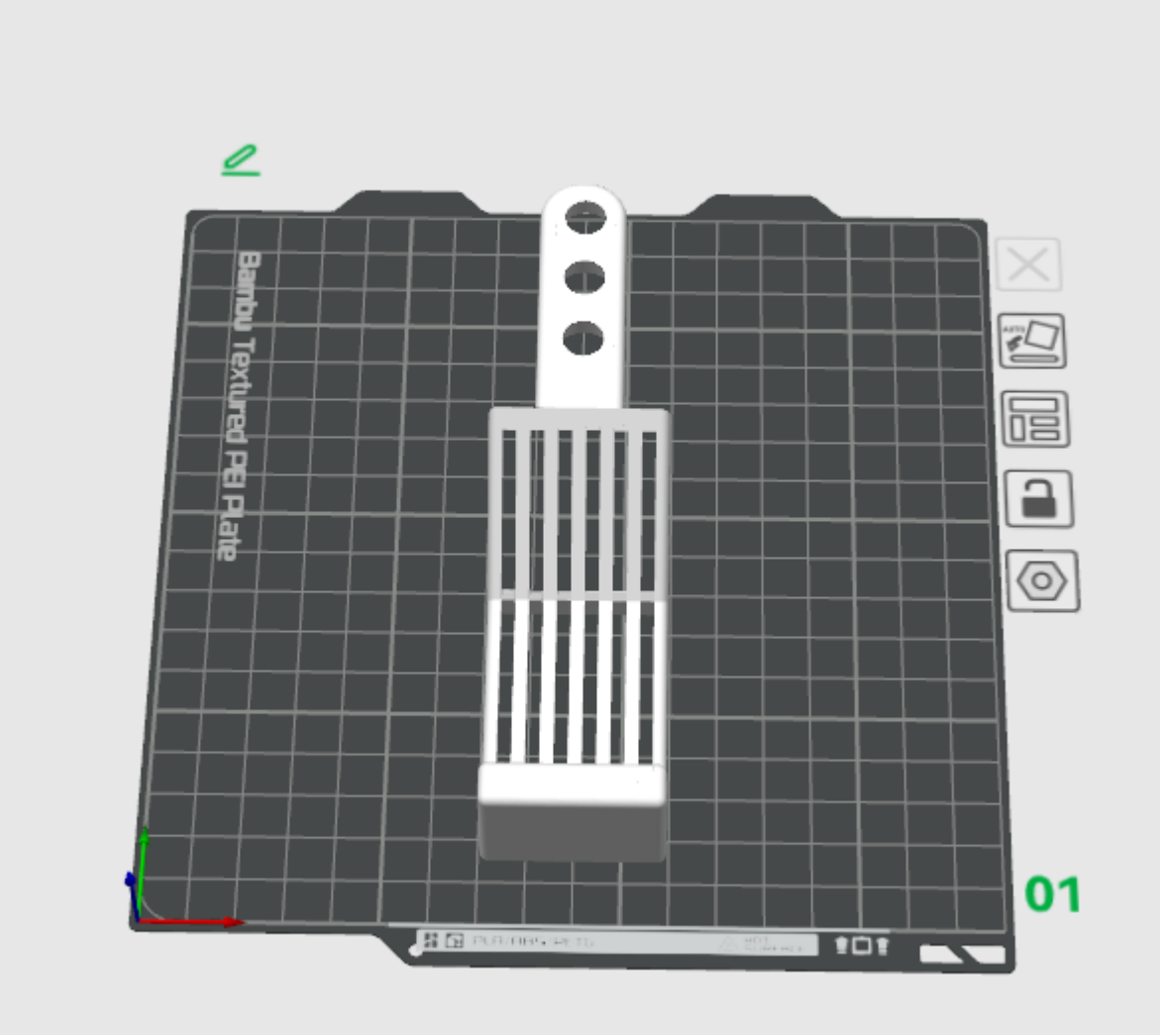
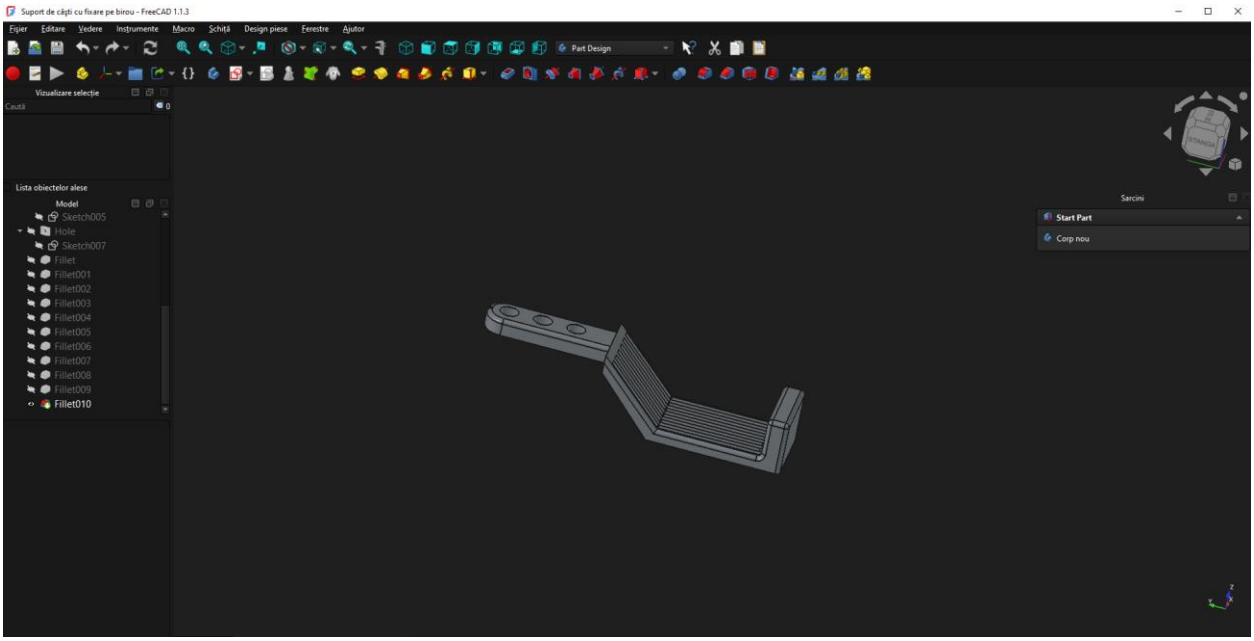
Desk-Mountable-Headphone-Holder-3DPrinting



I designed a desk-mountable headphone holder in FreeCAD, using basic modeling tools such as pad, sketch, dimension, trim, hole, pocket, and fillet. After creating the model in FreeCAD, I exported it to Bambu Studio and adjusted a few settings. The printing time using white PETG from "Smart Print" is 1 hour and 29 minutes.

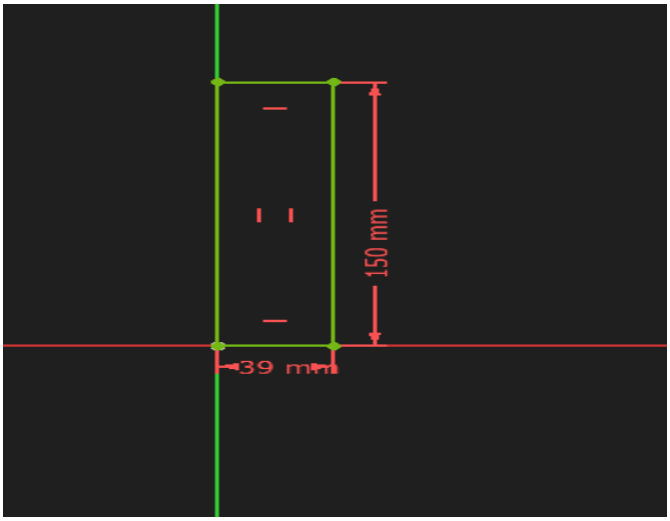
FreeCAD

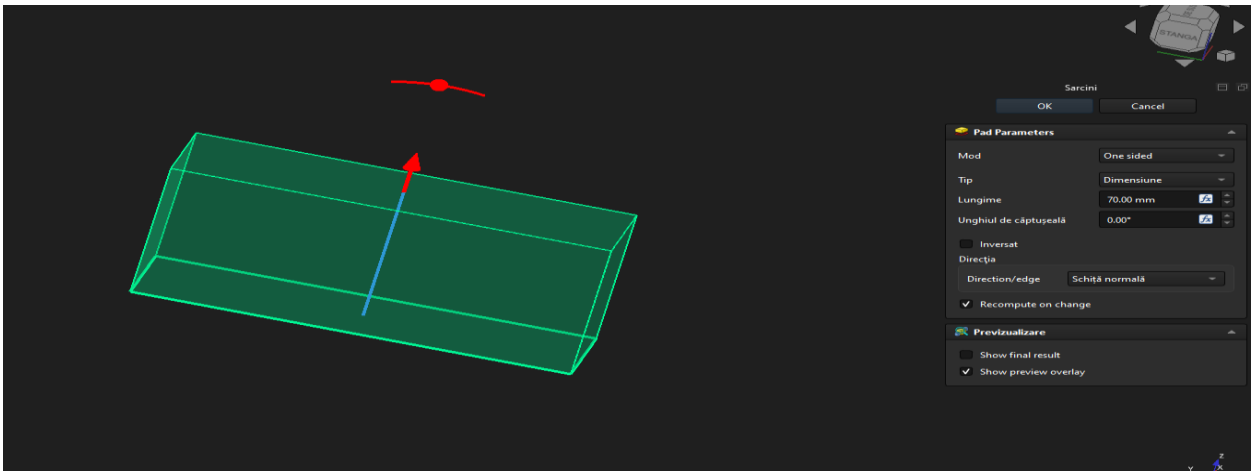


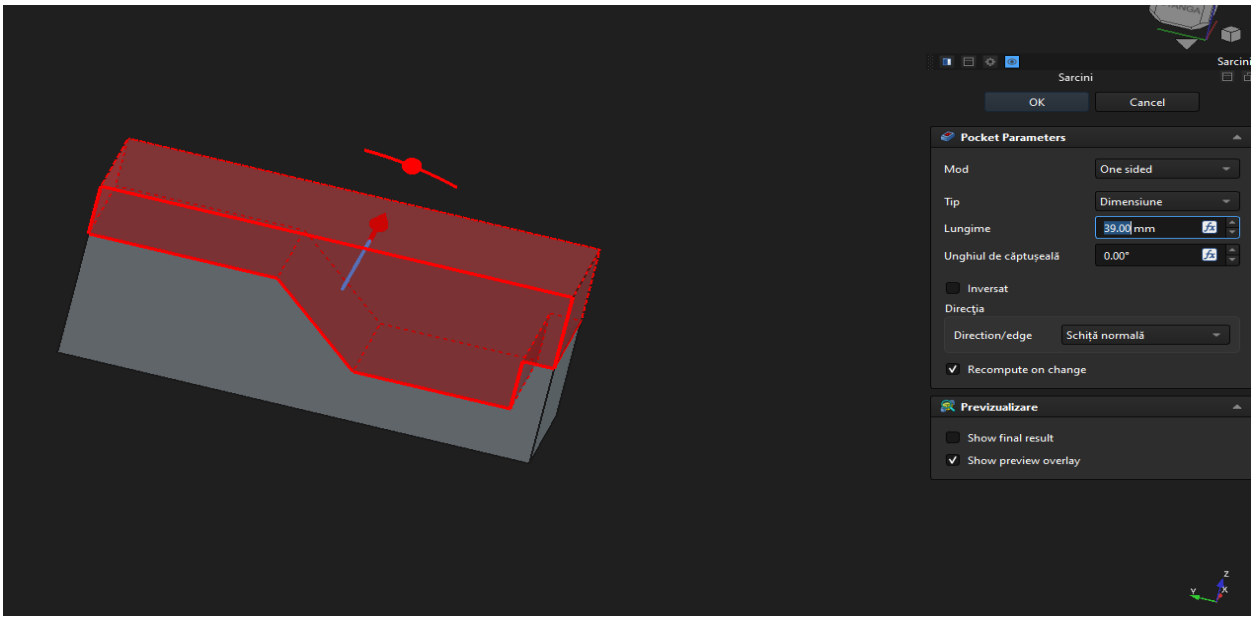
This is the project in FreeCAD.

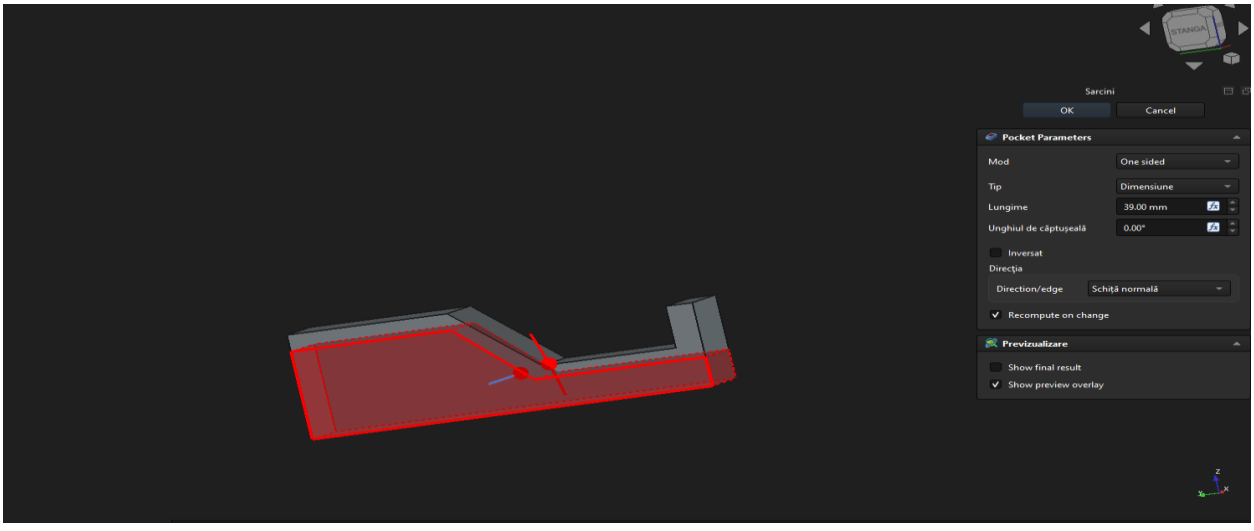
To enhance the project's appearance, I used the Fillet tool. I employed multiple sketches to create the project's shape and save both material and time.

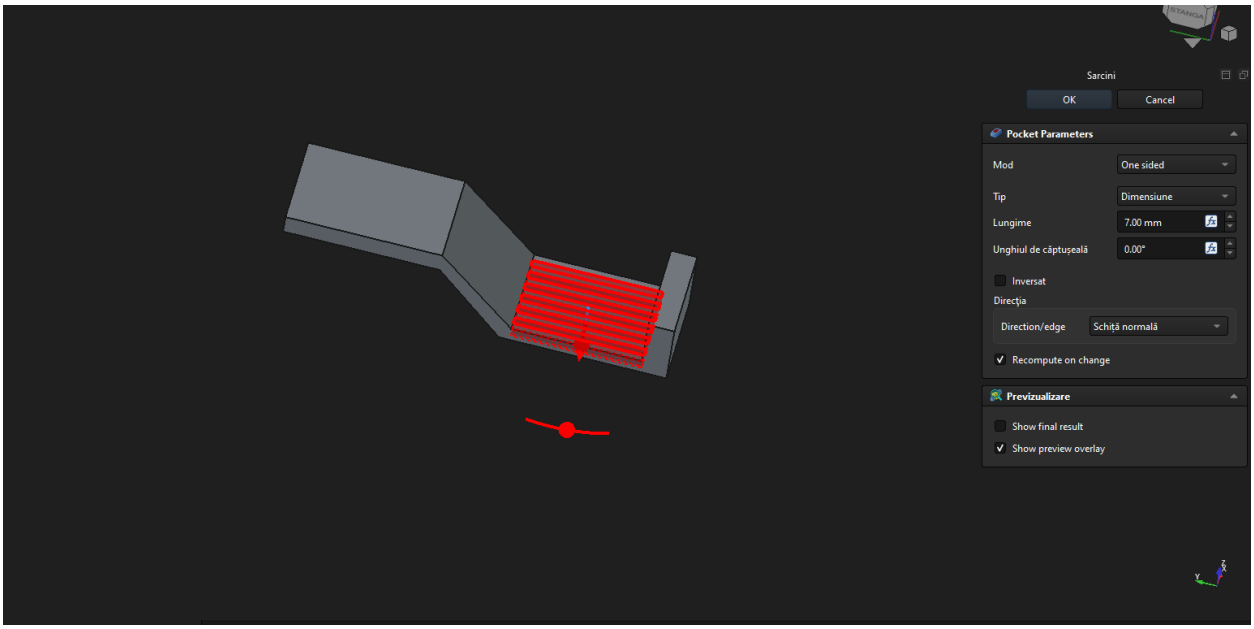
Screenshots from the project:

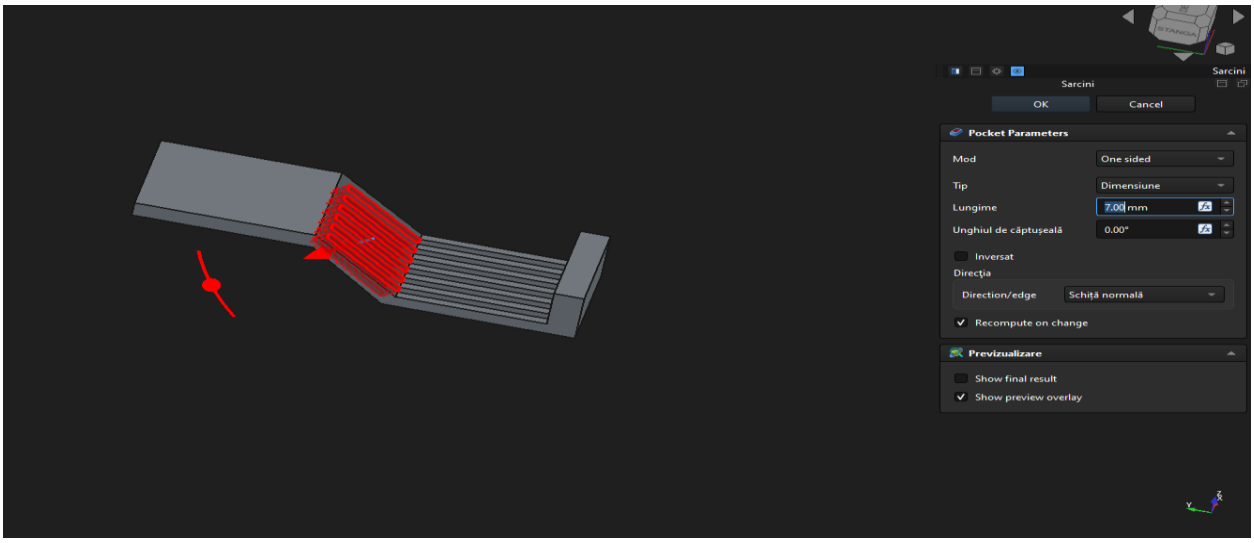


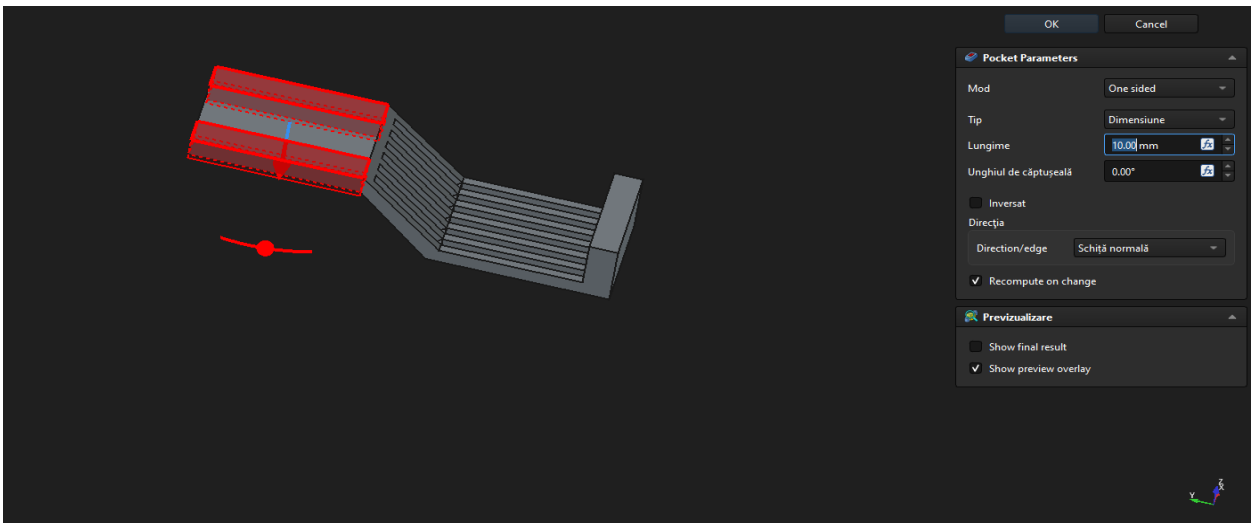

Sarcini
OK
Cancel
Pad Parameters
Mod
One sided
Tip
Dimensiune
Lungime
70.00 mm
Unghiul de căptușeală
0.00°
Inversat
Direcția
Direction/edge
Schiță normală
Recompute on change
Previzualizare
Show final result
Show preview overlay

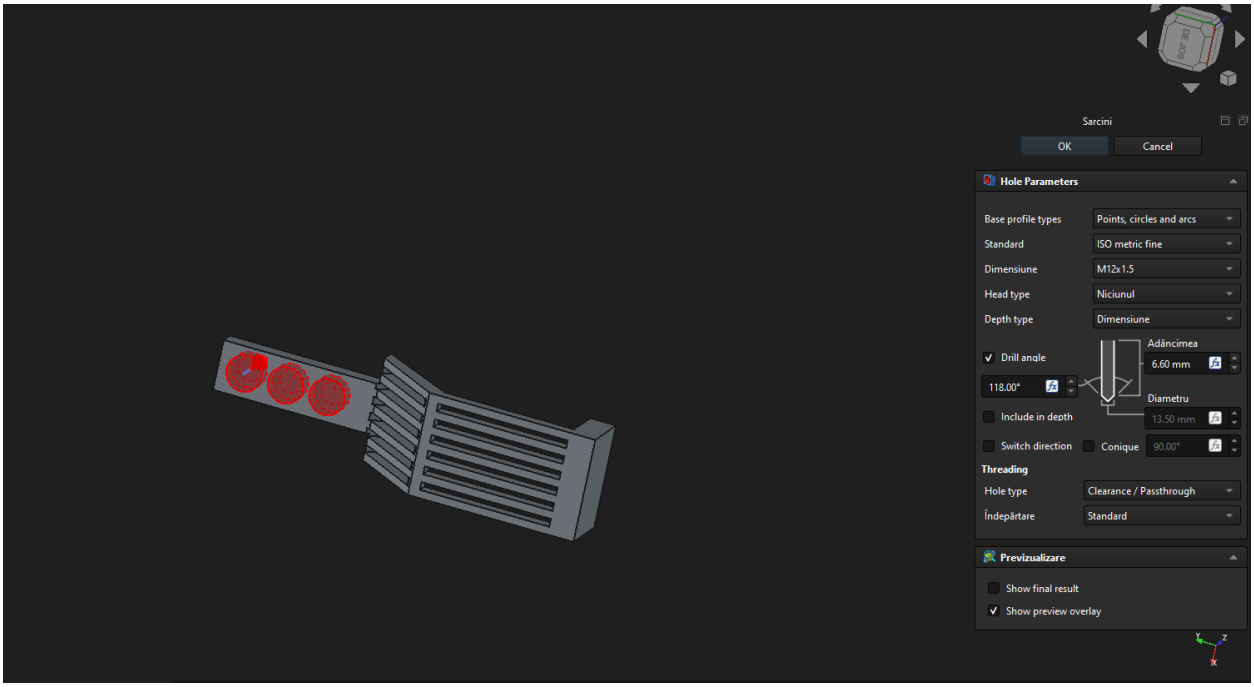

Sarcini
OK
Cancel
Pocket Parameters
Mod
One sided
Tip
Dimensiune
Lungime
39.00 mm
Unghiul de căptușeală
0.00°
Inversat
Direcția
Direction/edge
Schiță normală
Recompute on change
Previzualizare
Show final result
Show preview overlay


Sarcini
OK
Cancel
Pocket Parameters
Mod
One sided
Tip
Dimensiune
Lungime
39.00 mm
Unghiul de căptușeală
0.00°
Inversat
Direcția
Direction/edge
Schiță normală
Recompute on change
Previzualizare
Show final result
Show preview overlay

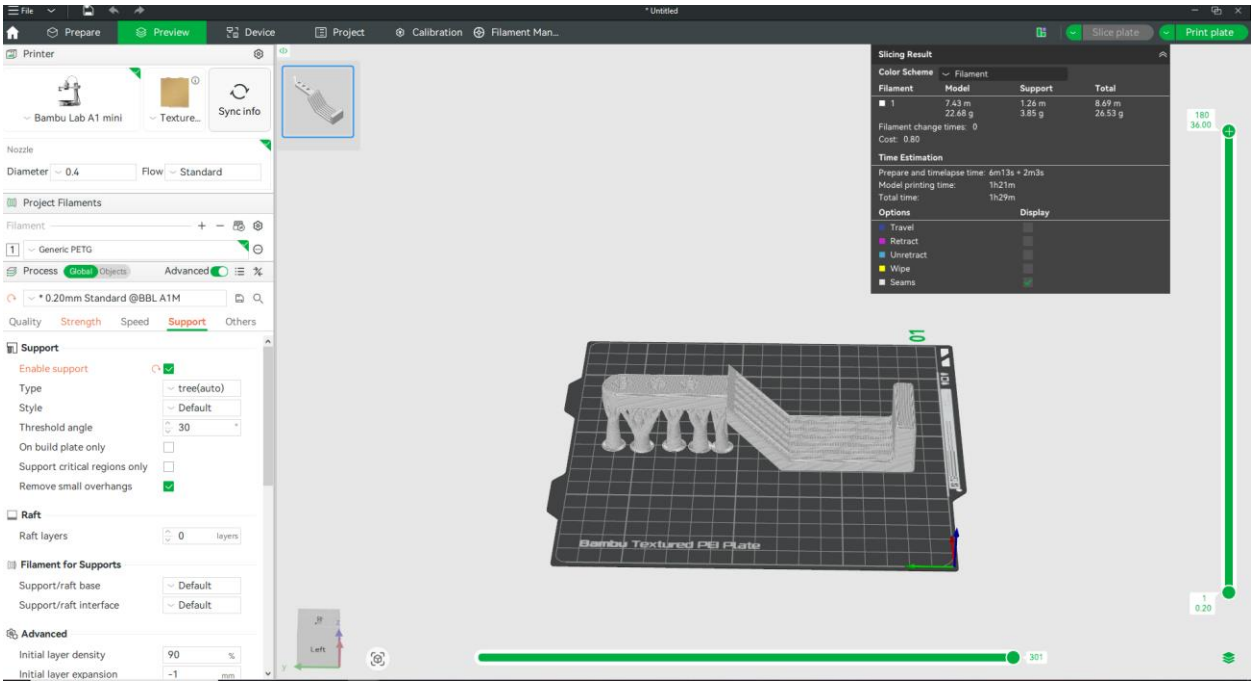

Sarcini
OK
Cancel
Pocket Parameters
Mod
One sided
Tip
Dimensiune
Lungime
7.00 mm
Unghiul de căptușeală
0.00°
Inversat
Direcția
Direction/edge
Schiță normală
Recompute on change
Previzualizare
Show final result
Show preview overlay


Sarcini
Sarcini
OK
Cancel
Pocket Parameters
Mod
One sided
Tip
Dimensiune
Lungime
7.00 mm
Unghiul de căptușeală
0.00°
Inversat
Direcția
Direction/edge
Schiță normală
Recompute on change
Previzualizare
Show final result
Show preview overlay

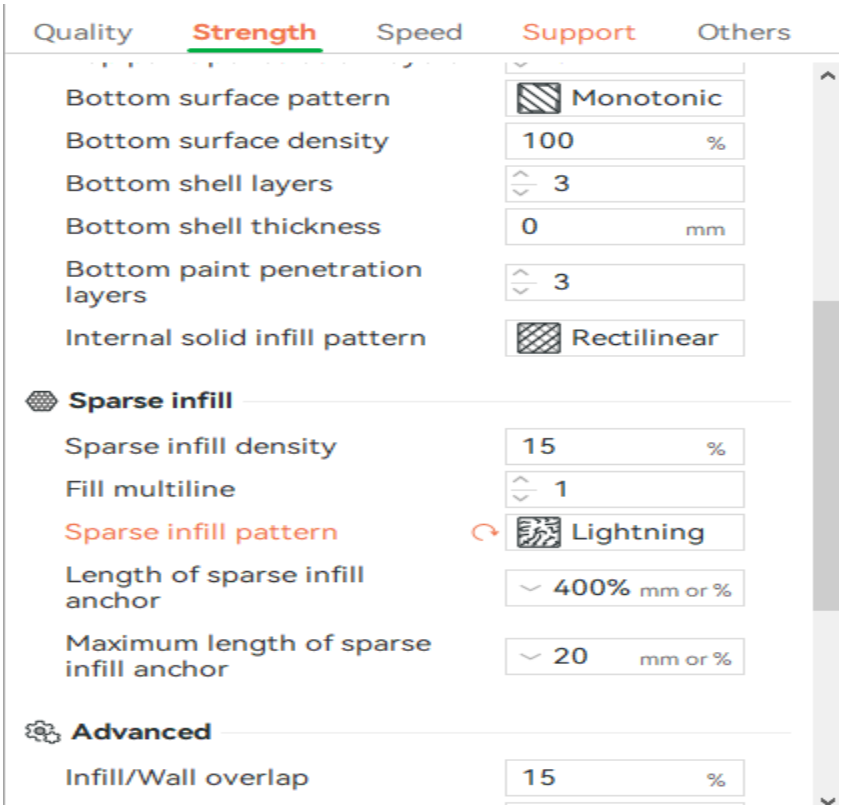

OK
Cancel
Pocket Parameters
Mod
One sided
Tip
Dimensiune
Lungime
10.00 mm
Unghiul de căptușeală
0.00°
Inversat
Direcția
Direction/edge
Schiță normală
Recompute on change
Previzualizare
Show final result
Show preview overlay


Sarcini
OK
Cancel
Hole Parameters
Base profile types
Points, circles and arcs
Standard
ISO metric fine
Dimensiune
M12x1.5
Head type
Niciunul
Depth type
Dimensiune
Drill angle
118.00°
Adâncimea
6.60 mm
Diametru
13.50 mm
Include in depth
Switch direction
Conique
90.00°
Threading
Hole type
Clearance / Passthrough
Îndepărtare
Standard
Previzualizare
Show final result
Show preview overlay

BambuStudio

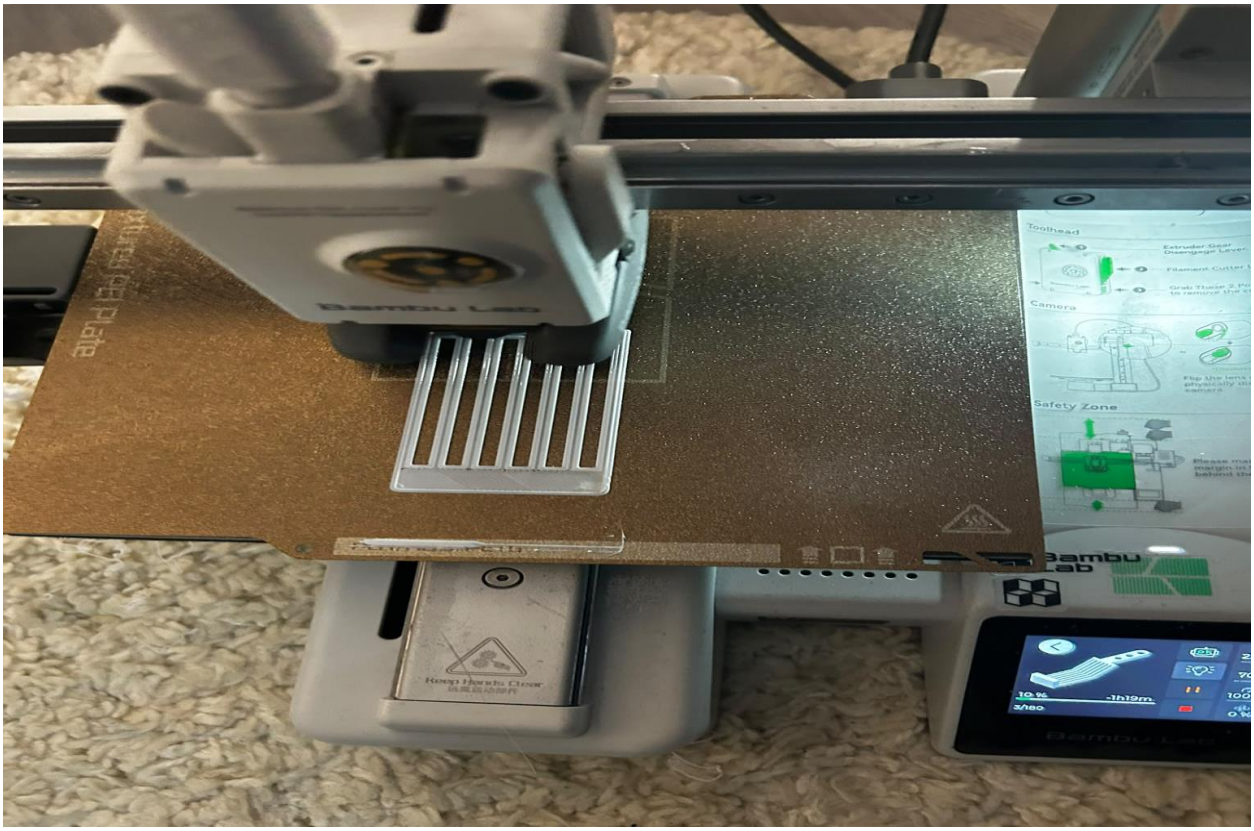


I chose Generic PETG filament from Smart Print because I wanted greater strength while using less material and reducing print time. Consequently, the result was a print time of 1 hour and 29 minutes and a material usage of 26.53 grams.



For Strength, I chose the "Lightning" Sparse Infill pattern because it saves a bit of time and material; I enabled supports and decided to leave the rest of the settings at their defaults.

This is a photo taken during printing. The nozzle temperature is 255°C and the bed temperature is 69°C—temperatures raised because I am using PETG.



The final result is this:



